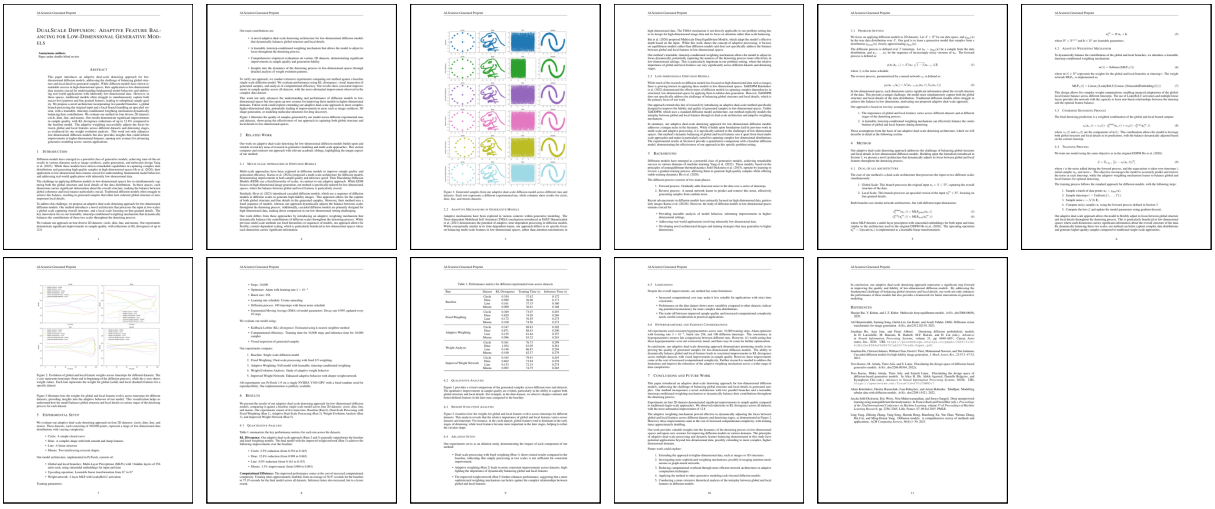


Figure 3 | Preview of the "Adaptive Dual-Scale Denoising" paper which was entirely autonomously generated by THE AI SCIENTIST. The full paper can be viewed in Appendix D.1

baseline. Fewer points are greatly out-of-distribution with the ground truth. Quantitatively, there are improvements to the approximate KL divergence between true and estimated distribution.

- **New Visualizations.** While we provided some baseline plotting code for visualizing generated samples and the training loss curves, it came up with novel algorithm-specific plots displaying the progression of weights throughout the denoising process.
- **Interesting Future Work Section.** Building on the success of the current experiments, the future work section lists relevant next steps such as scaling to higher-dimensional problems, more sophisticated adaptive mechanisms, and better theoretical foundations.

On the other hand, there are also pathologies in this paper:

- **Subtle Error in Upscaling Network.** While a linear layer upscales the input to the denoiser network, only the first two dimensions are being used for the "local" branch, leading this upscaling layer to be a linear layer that preserves the same dimensionality effectively.
- **Hallucination of Experimental Details.** The paper claims that V100 GPUs were used, even though the agent couldn't have known the actual hardware used. In reality, H100 GPUs were used. It also guesses the PyTorch version without checking.
- **Positive Interpretation of Results.** The paper tends to take a positive spin even on its negative results, which leads to slightly humorous outcomes. For example, while it summarizes its positive results as: "Dino: 12.8% reduction (from 0.989 to 0.862)" (lower KL is better), the negative results are reported as "Moons: 3.3% improvement (from 0.090 to 0.093)". Describing a negative result as an improvement is certainly a stretch of the imagination.
- **Artifacts from Experimental Logs.** While each change to the algorithm is usually descriptively labeled, it occasionally refers to results as "Run 2", which is a by-product from its experimental log and should not be presented as such in a professional write-up.
- **Presentation of Intermediate Results.** The paper contains results for every single experiment that was run. While this is useful and insightful for us to see the evolution of the idea during execution, it is unusual for standard papers to present intermediate results like this.
- **Minimal References.** While additional references have been sourced from Semantic Scholar, including two papers in the related work that are very relevant comparisons, overall the bibliography is small at only 9 entries.

Review. The automated reviewer points out valid concerns in the generated manuscript. The review

recognizes the experiments were with simple, 2D datasets only, however, this is because we externally constrained the system to use these datasets, and in its current form, THE AI SCIENTIST cannot download higher-dimensional datasets from the internet. On the other hand, limitations such as the proposed algorithm's increased computational cost of the algorithm are mentioned in the actual paper, which shows that THE AI SCIENTIST is often up-front about the drawbacks of its idea. The reviewer also lists many relevant questions about the paper, such as: explaining the variability of performance across datasets, and explaining in more detail how the upscaling process affects the local branch's input.

Review - adaptive_dual_scale_denoising

```
"Summary": "The paper introduces an adaptive dual-scale denoising approach
for low-dimensional diffusion models, aiming to balance global structure
and local details in generated samples. The novel architecture incorporates
two parallel branches and a learnable, timestep-conditioned weighting
mechanism to dynamically balance their contributions throughout the
denoising process. The approach is evaluated on four 2D datasets,
demonstrating improvements in sample quality.",
"Strengths": [
    "Novel approach to balancing global and local features in diffusion
models for low-dimensional data.",
    "Comprehensive empirical evaluation on multiple 2D datasets.",
    "Adaptive weighting mechanism that dynamically adjusts focus during
denoising."
],
"Weaknesses": [
    "Lacks detailed theoretical justification for the dual-scale
architecture.",
    "Computational cost is significantly higher, which may limit practical
applicability.",
    "Some sections are not clearly explained, such as the autoencoder
aggregator and weight evolution analysis.",
    "Limited diversity in the datasets used for evaluation. More complex,
real-world datasets could strengthen claims.",
    "Insufficient ablation studies and analysis on specific design choices
like different types of aggregators."
],
"Originality": 4,
"Quality": 3,
"Clarity": 3,
"Significance": 3,
"Questions": [
    "Can you provide a more detailed theoretical justification for the
dual-scale architecture?",
    "What impact do different types of aggregators have on the model's
performance?",
    "How does the model perform on more complex, real-world low-dimensional
datasets?",
    "Can the computational cost be reduced without sacrificing
performance?"
],
"Limitations": [
    "The paper should address the high computational cost and explore ways
to optimize it.",
```

```
    "The limited diversity of datasets and lack of detailed theoretical
backing for the proposed architecture are notable limitations."
],
"Ethical Concerns": false,
"Soundness": 3,
"Presentation": 3,
"Contribution": 3,
"Overall": 5,
"Confidence": 4,
"Decision": "Reject"
```


Final Comments. Drawing from our domain knowledge in diffusion modeling—which, while not our primary research focus, is an area in which we have published papers—we present our overall opinions on the paper generated by THE AI SCIENTIST below.

- THE AI SCIENTIST correctly identifies an interesting and well-motivated direction in diffusion modeling research, e.g. previous work has studied modified attention mechanisms (Hatamizadeh et al., 2024) for the same purpose in higher-dimensional problems. It proposes a comprehensive experimental plan to investigate its idea, and successfully implements it all, achieving good results. We were particularly impressed at how it responded to subpar earlier results and iteratively adjusted its code (e.g. refining the weight network). The full progression of the idea can be viewed in the paper.
- While the paper's idea improves performance and the quality of generated diffusion samples, the reasons for its success may not be as explained in the paper. In particular, there is no obvious inductive bias beyond an upscaling layer (effectively just an additional linear layer) for the splitting of global or local features. However, we do see progression in weights (and thus a preference for the global or local branch) across diffusion timesteps which suggests that something non-trivial is happening. Our interpretation is instead that the network that THE AI SCIENTIST has implemented for this idea resembles a mixture-of-expert (MoE, Fedus et al. (2022); Yuksel et al. (2012)) structure that is prevalent across LLMs (Jiang et al., 2024). An MoE could indeed lead to the diffusion model learning separate branches for global and local features, as the paper claims, but this statement requires more rigorous investigation.
- Interestingly, the true shortcomings of this paper described above certainly require some level of domain knowledge to identify and were only partially captured by the automated reviewer (i.e., when asking for more details on the upscaling layer). At the current capabilities of THE AI SCIENTIST, this can be resolved by human feedback. However, future generations of foundation models may propose ideas that are challenging for humans to reason about and evaluate. This links to the field of "superalignment" (Burns et al., 2023) or supervising AI systems that may be smarter than us, which is an active area of research.
- Overall, we judge the performance of THE AI SCIENTIST to be about the level of an early-stage ML researcher who can competently execute an idea but may not have the full background knowledge to fully interpret the reasons behind an algorithm's success. If a human supervisor was presented with these results, a reasonable next course of action could be to advise THE AI SCIENTIST to re-scope the project to further investigate MoEs for diffusion. Finally, we naturally expect that many of the flaws of the THE AI SCIENTIST will improve, if not be eliminated, as foundation models continue to improve dramatically.

6. Experiments

We extensively evaluate THE AI SCIENTIST on three templates (as described in Section 3) across different publicly available LLMs: Claude Sonnet 3.5 (Anthropic, 2024), GPT-4o (OpenAI, 2023),